

1. Differences between HTTP and HTTPS

HTTP (HyperText Transfer Protocol) transfers data in plain text — anyone can intercept and read it.

HTTPS (HTTP Secure) adds SSL/TLS encryption, making the data unreadable to eavesdroppers.

HTTPS requires an SSL certificate and uses port 443, while HTTP uses port 80. All websites handling sensitive data (login, banking, APIs) must use HTTPS.

2. Structure of an HTTP Request and Response

HTTP Request:

```
GET /api/users HTTP/1.1
Host: example.com
Accept: application/json
Authorization: Bearer mytoken123
```

HTTP Response:

```
HTTP/1.1 200 OK
Content-Type: application/json

{"id": 1, "name": "Alice"}
```

3. Common HTTP Methods

- **GET** — Retrieves data. Use case: Fetching a list of users from an API.
 - **POST** — Creates a new resource. Use case: Registering a new account.
 - **PUT** — Replaces an existing resource completely. Use case: Updating a full user profile.
 - **PATCH** — Partially updates a resource. Use case: Changing only a user's email.
 - **DELETE** — Removes a resource. Use case: Deleting a user account.
-

4. Common HTTP Status Codes

- **200 OK** — Request succeeded. Scenario: GET /users returns the list successfully.
- **201 Created** — Resource created. Scenario: POST /users creates a new account.
- **400 Bad Request** — Invalid request syntax. Scenario: Sending malformed JSON in the body.
- **401 Unauthorized** — Authentication required. Scenario: Accessing /profile without a token.
- **403 Forbidden** — Authenticated but not allowed. Scenario: A regular user tries to access an admin route.
- **404 Not Found** — Resource doesn't exist. Scenario: GET /users/9999 when that user doesn't exist.
- **500 Internal Server Error** — Server-side bug. Scenario: A crash in the API code when processing a request.